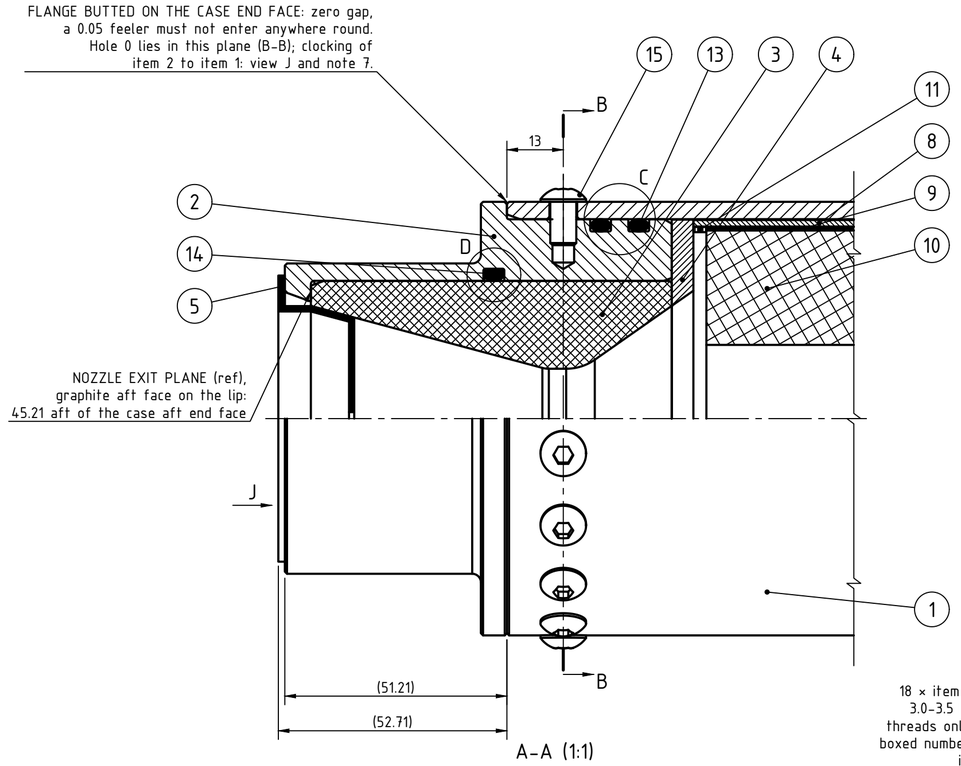
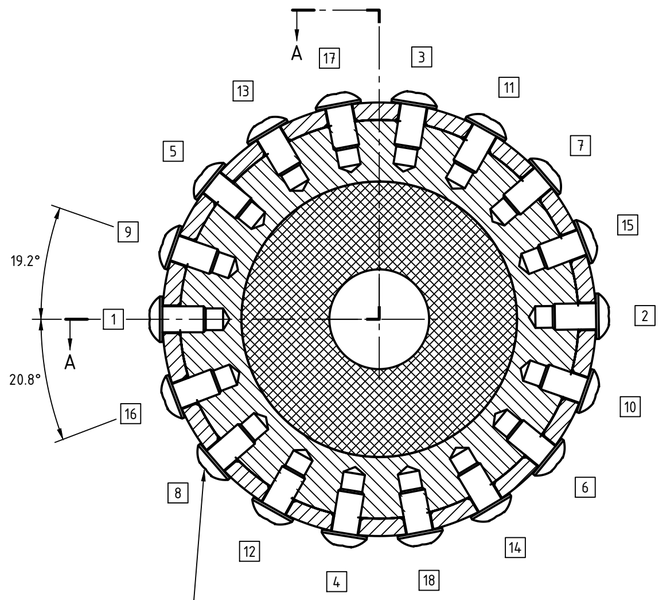
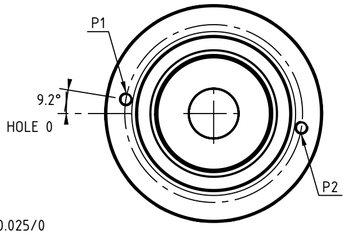
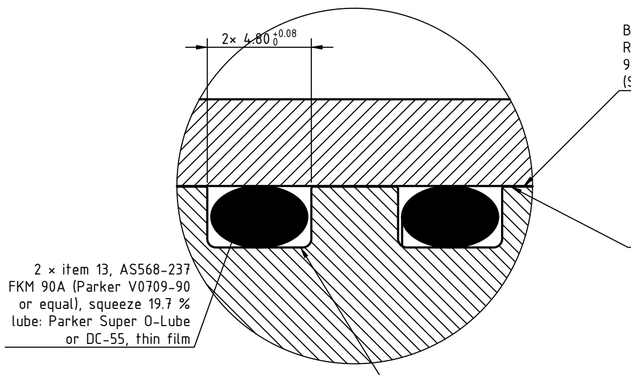




18 × item 15, M6×10–010.9 (ISO 7380-1) 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Order: the boxed numbers, 3 passes (note 8). Holes in item 1: ⌀6.40 +0.03/0 (ref)



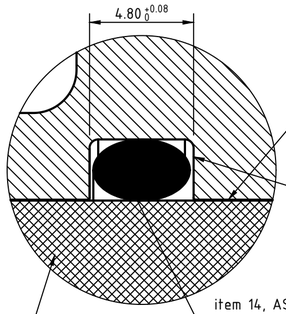
At entry, seen from aft: hole 0 of item 2 on hole 0 of item 1, clocking line straight across both, AFT markings side by side (note 7). Hole 0 is 9.2° counter-clockwise of puller P1 (θ 10°); P2 at θ 190°, both on r 41: 2 × M6×1–6H ↧8, ⌀5 ↧10 (SRM100-102).



BORE, item 1: ⌀A +0.030/0, honed Ra 0.4 (0.8 max), circumferential lay; 92.000 ≤ A ≤ 92.070, recorded per zone (SRM100-101). Drawn at A = 92.000.

LAND, item 2: ⌀A – 0.06 0/–0.02 (⌀91.94 at A = 92.000) clearance on ⌀ 0.06–0.11 (0.030–0.055 radial)

2 GROOVES ⌀A – 5.67 0/–0.04 (⌀86.33 at A = 92.000) gland depth 2.835 (ref)



BORE, item 2: ⌀63.58 +0.025/0
INSERT, item 3: ⌀63.50 0/–0.05
clearance on ⌀ 0.080–0.155

GROOVE ⌀69.14 +0.05/0 at z –3 gland depth 2.820 (ref)

ITEM 3: slide in by hand, aft end first. Never press or knock it in. Push sleeve: extraction only (note 13).

item 14, AS568-230, FKM 90A (Parker V0709-90 or equal), squeeze 20.1 %, lube as item 13, fitted in item 2's groove first

AXIAL STATIONS, z from the case aft end face, + forward

STATION (ref)	z
Cap flange back face (motor aft end)	–52.71
Holder aft face	–51.21
Nozzle exit plane (graphite aft face)	–45.21
Holder flange aft face	–6.00
Insert O-ring groove, centre	–3.00
Case aft end face = flange face (datum)	0.00
Bolt row, 18 × item 15	13.00
Case O-ring grooves, centres	21.60 / 30.40
Holder and insert forward faces	38.00
Insulator fwd face = liner and spacer 0 aft	43.00
Grain 1 aft face	46.00

AFT END BUILD-UP AND INSTALLATION

1. Item numbers, materials and part numbers: parts list on sheet 1. Tooling named here (hole plugs, push sleeve, puller bridge) is not part of this drawing set.
2. Before the first build (README, case): item 1 per SRM100-101, each seal zone honed to the smallest diameter that cleans up, A (the smallest reading in the zone, 92.000 ≤ A ≤ 92.070), held to A +0.030/0 and recorded per zone; items 2 and 6 cut to their zone's A; each end match-drilled with its closure, serialised and marked (SRM100-101 note 8, SRM100-102 note 8, SRM100-103 note 11), clocking line scribed; hole baseline recorded (SRM100-101 R7). Then the hydrostatic proof (SRM100-001, proof dummy SRM100-120 in place of item 3, no insulators, liner or propellant): before the first firing and after any repair, any retired-and-replaced part or a label reading ≥ 232 °C (note 12).
3. The parts replaced after every firing (note 14) are always new. Lubricant for every O-ring, the liner OD and the insulator edges: Parker Super O-Lube or DC-55, thin film.
4. Fit item 14 in the bore groove of item 2, then 2 × item 13 in the grooves of its seal land (details C, D). Item 2 is cut to the aft-zone bore A recorded on SRM100-101 (92.000 ≤ A ≤ 92.070): land A – 0.06 0/–0.02, groove bottoms A – 5.67 0/–0.04 (91.94 / 86.33 at A = 92.000). Details C and D are drawn at A = 92.000.
5. Slide insert item 3 into item 2 by hand, aft (exit) end first, until it seats on the lip. Never press or knock it. Check: its forward face lies flush with item 2's forward face to 0.10 below it (seat 83.21 +0.05/0, insert 83.21 0/–0.05). Then seat nozzle insulator item 4 on item 2's forward face.
6. Hole plugs: push 18 flat-ended acetal (Delrin) plugs, ⌀6.35 × 3.85 with an outside head, into the case's aft-end holes. Their ends sit about 0.15 mm shy of the bore, so no O-ring is dragged across an open hole edge.
7. Clocking (view J): enter item 2 with its hole 0 on the case's hole 0, the scribed clocking line straight across the case end and the flange and the AFT markings of items 1 and 2 (SRM100-101 note 8, SRM100-102 note 8) side by side. Check: seen from aft, hole 0 is 9.2° counter-clockwise of puller P1 (θ 10°). Push in by hand until the flange butts on the case end face (zero gap).
8. Pull each plug as its screw goes in. Screws 18 × item 15, 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Tighten in the boxed order of section B–B (opposite pairs, criss-cross): pass 1 to about 1.5 N·m, pass 2 to final torque, pass 3 at final torque hole by hole round the circle, as a check. Never fit an M6×16 (item 16) here: it bottoms with its head proud.
9. Load the liner and grain stack and close the forward end: sheet 3.
10. Temperature labels, new for every firing: irreversible 177 / 204 / 232 / 260 °C strips on item 1's OD at z 20–35, θ 90° and 270° (SRM100-101 note 8), and on item 2's aft body ⌀71.58 at θ 90° and 270° (SRM100-102 note 8), clear of the cap tape (note 11).
11. Cap item 5: fit it now, in the exit, flange on item 2's aft face, taped per SRM100-106 note 6. It is the debris seal and stays on through storage and transport. At the stand it comes off only to insert the igniter, which always goes in last (leads shunted, out through the cap's ⌀3 hole); then it is refitted and re-taped with new tape (sheet 1 note 4). Never move the motor with an igniter in it.
12. After every firing, before extraction: read and record the temperature labels on items 1 and 2 (note 10). A reading ≥ 232 °C: retire or re-proof that end (SRM100-101 R10, SRM100-001).
13. Extraction, never lever or knock. Closure: split clamp collar (100.0 bore) on the case OD at least 50 mm from the end, clear of the screw heads at z 13 (sweep ⌀106.04 ref), the labels and the marking at z 20–45; puller bridge and 2 × M6 draw screws in item 2's puller threads P1, P2 (view J). Insert: with item 2 out, support it on its forward face in an arbor press and push item 3 out forward with the push sleeve (⌀52.4 ID × ⌀53.8 OD) through the lip bore onto its push face, r 26.2–27.0. The push sleeve is for extraction only.
14. Then, after every firing: measure the throat. Replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels. Re-use items 1, 2 and 6 after inspection (SRM100-101 R10 for the case), and item 3 while its throat is ≤ ⌀23.9 and it is uncracked. On the aft case end, measure hardness and conductivity again and re-gauge every bolt hole against its baseline (SRM100-101 R10).

USED ON: SRM-100 motor (Helios) – static test and flight

MATERIAL: See parts list, sheet 1	MASS: 10.60 kg	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: –	EDGES: –
DOCUMENT TYPE: ASSEMBLY DRAWING	DOCUMENT STATUS: NOT CHECKED – FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: MOTOR ASSEMBLY – AFT END — Aft closure build-up and installation, seal glands, tightening sequence			
CREATED BY: text-to-cad / Claude	APPROVED BY: –	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-000	REV: A
				SHEET: 2 / 3